Effective Modeling and Visualization Strategies for Major Changes within the College of Engineering and Applied Science at the University of Colorado Boulder

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Abstract: College students frequently switch majors and The University of Colorado Boulder (CU Boulder) College of Engineering and Applied Science (CEAS) seeks to understand these trends in major switching within their college in order to properly allocate resources in light of unpredictable enrollment numbers. Two strategies are deployed to bring light to these patterns. First, a hierarchical model is fit to represent the probability of major switching throughout time. Second, an app is developed which incorporates three interactive visualization panels, showing students' major transition patterns relative to a specific major of interest. The hierarchical model found that lower term GPA is a predictor of higher major switching probability. The model also identified certain programs with higher and lower relative propensities for major switching. The visualization software utilized alluvium plots, network diagrams, and bar-charts, allowing for investigation of student transition data on various scales of analysis. The interpretable and interactive nature of the visualizations will allow the administrators of the CEAS to answer a wide variety of questions pertaining to student transitions, including common academic trajectories between majors, department retention or growth, and the typical academic histories of students who switch majors.

I. INTRODUCTION AND BACKGROUND

The University of Colorado Boulder (CU Boulder) contains seven undergraduate colleges and over 50 majors. The College of Engineering and Applied Science (CEAS) admitted 831 students for the 2024–2025 academic year and has a total of 6,700 students currently enrolled [9]. Adequately assessing term-by-term major enrollment within the college is valuable for proper allocation of university resources, however, the variability in major enrollment is not well understood, and thus this prediction has historically been difficult to make.

A. Background

The CEAS has noticed a large number of students transferring between majors within the college throughout their time at CU Boulder. Additionally, the CEAS sees many Intra-University Transfer (IUT) enrollments. These are students that were enrolled in another college at CU Boulder (e.g., College of Exploratory Studies) before transferring into the College of Engineering and Applied Science. The CEAS has identified that these multiple enrollment sources contribute to the difficulty in predicting the number of students in each discipline. Nevertheless, the CEAS has yet to establish a clear way to model changes between engineering majors. Current CEAS software which visualizes term-by-term major enrollment data is difficult to use and is largely ineffective at communicating meaningful trends. Currently, administrators rely on this software and their heuristics to estimate program enrollment in their decision making process, which is not always effective. For example, in the Fall 2025 term, a notable underestimation of sophomore-to-junior level engineering students transferring into the Mechanical Engineering (MCEN) major presented challenges for professors due to the unexpectedly varied background knowledge in junior level MCEN students as well as underestimation of course enrollment demand leading to underestimation of course size. Adequate allocation of academic resources and understanding of common academic pathways is important to encourage student success and reduce administrative burden.

The CEAS also wants to support students in finding their path to success at CU, which includes the identification of students who are most likely to switch majors or leave the college and who may need support in deciphering their academic direction. The CEAS has not yet implemented a statistics-based workflow identifying student factors which predict switching majors, making the direction of these interventions difficult.

B. Literature Review

Major switching within STEM fields has been a central focus of higher education research, as understanding the reasons behind persistence and attrition is crucial for improving student retention. It has been found that most students leaving STEM programs transition to non-STEM majors rather than switching within STEM [1]. Building on this work, other studies have focused on retention in STEM programs, particularly the characteristics associated with students' decisions to remain in or leave STEM programs. For example, important differences by engineering discipline have been previously found: Industrial engineering appears to retain students more effectively by fostering a supportive environment, while chemical and electrical Engineering experience higher attrition despite similar grade patterns. Mechanical engineering attracts many students who will switch majors, but shows lower dropout rates despite grade patterns comparable to electrical engineering. These findings suggest that while disciplinary cultures influence retention patterns, individual academic performance is also a critical factor in students' decisions to remain in or leave STEM fields [4].

Academic performance has consistently been shown to influence major retention. First year or early college grade point averages (GPA) have been found to be a significant predictor in whether a STEM student switches majors [2, 4]. One study found that a one point decrease in GPA was associated with a 17% to 19% increase in the probability of a student leaving STEM [2]. Students who stayed in STEM were found to have the highest average early college GPAs [1]. Interestingly, over half of students who leave the CEAS maintain their GPAs, highlighting the importance of nonacademic factors such as advising, sense of belonging, and interest alignment [6]. Additionally, student persistence in engineering is strongly shaped by stronger high school preparation including GPA, math and science grades, and Advanced Placement (AP) STEM test scores. It has been found that higher SAT math scores predict flexible switching within engineering, whereas high SAT verbal scores correlate with leaving engineering entirely [4]. Furthermore, studies have shown that gateway courses matter. Performance in early math and science (calculus and introductory physics in particular) emerges as a pivotal filter, where D/F/W marks or low GPAs reduce persistence [6]. Additionally, some studies have shown demographic factors such as ethnicity and gender have a greater effect than academic performance within different demographic groups [3, 6]. It has been found that women switch from STEM majors primarily because of preferential reasons (e.g., to leave a male-dominated field) instead of academic ability, while men usually leave STEM due to lower grades [1, 6]. Understanding demographic factors in addition to academic performance may be useful in determining which students are at risk of leaving engineering majors.

Some effort has been made to model retention and the probability of switching majors. Models attempting to predict the probability of switching majors included course grades and student preference which may vary depending on demographic factors [2–4]. For instance, a regression model developed by Figari and Speer on the probability of a student switching majors focused on mismatches between academic ability and student preferences, with low grades being an indicator of an academic mismatch [2]. In contrast, the models developed by Chong Ho Yu et al. focused on retention and included student demographics and high school academic performance such as SAT scores. These models used data on the student population of Arizona State University as a whole and left out college academic performance as variables [7]. However, their multivariate adaptive regression splines (MARS) model found how many online class hours a student took during their sophomore year to be the most important predictor in retention. This reflects whether a student feels a sense of belonging within their major. Additionally, Main et al. developed a logistic regression model to predict the probability of students choosing a specific engineering major (e.g. chemical). Their model used demographic data, as well as high school and college academic performance, as variables. Main et al. found that demographic factors and academic performance were associated with a specific engineering major choice. For example, women were found to be more likely to major in chemical engineering and underrepresented minorities were more likely to major in electrical engineering. Furthermore, increases in first-year GPA in engineering courses were associated with decreases in the probability of majoring in civil and industrial engineering. These findings highlight that demographic and academic factors have different effects on the probability of majoring in specific majors and reveal differences in these factors depending on the engineering major. It is important to note that Main et al. focused on only five engineering majors: electrical, mechanical, civil, industrial, and chemical [8]. Nevertheless, these models provide important understanding of how the choice of various engineering majors differs depending on academic and demographic factors.

C. Knowledge Gap

Previous research has looked at other engineering programs and how their populations change over time; however, a similar analysis has not been performed within CU Boulder's CEAS specifically. With CU Boulder's IUT program and large engineering school, previous results may not generalize, an implementing similar models or visualizations will require additional considerations and data processing. Additionally, previous research lacks communicability, relying primarily on summary statistics to derive meaningful results. Efficacious use of the results of this paper within the CEAS are dependent upon the translation of statistical analyses into mediums interpretable by non-statistician administrators, who can then make informed administrative decisions based on data. This demands the usage of model architectures that are highly interpretable, or the creation of digestible and effective visualizations of student enrollment data.

D. Problem Statement

Up until this point, CU Boulder and the College of Engineering and Applied Science have yet to develop a comprehensive workflow to understand and predict the fluctuating numbers of students enrolled in each major within the college. The goal of this project is to create an effective and interpretable representation of changing major populations within the CEAS for administrative use, as well as to develop an understanding of the trends within students who switch between majors in the school and those who leave the school entirely. Ultimately, the tools we create for allocating information will be used to inform administrative decisions such as the allocation of campus resources and the direction of retention interventions.

Going forward, this project hopes to build models and visualizations that clearly depict the population flow between majors within the CEAS. Focusing on when majors see the largest influx or loss of students throughout the course of each program, the goal is to understand at which points in programs do students tend to switch in or out so that departments are more prepared and can adjust to fluctuating program sizes and needs.

II. METHODOLOGY

This section outlines the methodological framework used for the project. It begins by describing the software tools and programming environments utilized for data cleaning, wrangling, and visualization. Following this, the data set provided by the College of Engineering and Applied Science is introduced in detail, including its structure, key variables, and the transformations applied to prepare it for analysis.

A. Tools

For this project, different software was used depending on the task. For aggregating and cleaning the data, R version 4.5.1 and Microsoft Excel were used. Excel was used to primarily sort through and remove columns that were irrelevant to this project. R was the primary programming language used for cleaning and wrangling the data into an appropriate form to analyze. Tools from the ”tidyverse” package were used primarily to accomplish this task. In particular, this package was used to create a cleaned data set that could be used for developing models. For data analysis and visualization, R was used. The ”tidyverse” and ”dplyr” packages were used to analyze the data, the ”lme4” package was used for model development. For data visualizations we used several packages. The package ”ggplot2” was used to create the bar charts. The package ”ggalluvial” was used to create the alluvial charts. The package ”igraph” was used to create the network charts. The packages ”shiny” and ”shinyWidgets” were used to incorporate interactive visualizations into an app.

B. Data

The data was provided by CU Boulder CEAS and contained every student who had been enrolled in the college from the Fall 2013 term to Spring 2025. The data set consists of 21,007 rows, each representing a single student. For each student, 99 variables were recorded, including college application data, high school GPA, as well as the current college, major, cumulative GPA, and term GPA for up to 12 terms of enrollment.

Several new variables were created to facilitate data analysis. Although the data set included an indicator for whether a student graduated, it did not clearly distinguish between students who were still enrolled and those who had left CU Boulder. To address this, dropout/transfer status was inferred using a combination of the graduation indicator, the completeness of each student's major history, and their final term of enrollment. Using these components, a final status variable was constructed to classify every student as graduated, currently enrolled, or left the university.

Term codes stored in a numeric YYYY# format were converted into readable semester labels (e.g., “Spring 2022”). Several incomplete standardized test score variables with high occurrences of missing values were removed, as they contained little usable information.

Because many CEAS majors changed names during the period of data collection, all historical major codes were updated to their current acronyms (e.g., TCAM is a historical major code for TMEN). To further reduce noise, certain major codes were grouped together. If programs were run by the same department and were deemed functionally similar enough to each other, they were grouped together (e.g., Electrical Engineering (EEEN) and Electrical and Computer Engineering (ECEN) were grouped together as EEEN). The decision to group certain majors in this manner was made in collaboration with administrative faculty and was largely based on the similarity of the curriculum.

After cleaning and standardizing student records, the term-level variables (major, college, cumulative GPA, and term GPA across up to twelve terms) were reshaped from wide to long format so that each row represented a single student-term. This structure allowed term-by-term analysis of academic pathways. Within the long-format data, major and college switch indicators were created by comparing each term's values to those of the previous term.

C. Approach

This section outlines two complementary analytical approaches. The first uses visualizations of student movement between majors over time, highlighting overall trends in major switching relative to various academic factors. The second employs logistic regression models to predict individual switching behavior based on academic and demographic factors. While the visualizations provide a macro-level view of student flow, the model offers a micro-level perspective on the factors driving those transitions.

1. Visualization

Individual student academic trajectories were identified as the sequence of majors that a student has been enrolled in during their time at CU Boulder. Due to the high dimensionality of the observed trajectories ($n = 2205$), creating one plot to communicate the totality of the patterns within academic trajectories was infeasible. Instead, the prevalence and time course of specific academic trajectories was investigated through bar charts, flow diagrams, and alluvium plots.

These visualization methods have been automated and combined into one centralized app. This application allows administrators to create their own visualizations of student data according to the described plot types. For all plot types, the user must select a major of interest to visualize, then specify a variety of data filters used to target analysis on subsets of the student population.

2. Model

A multilevel logistic regression was created to predict the probability of a student switching majors in the CEAS in a given term. In this model, the slopes of the variables varied based on the number of terms a student took in order to introduce random effects, which made the slopes of the predictors dependent on the term. An indicator variable was created to determine whether a student switched majors within the CEAS which was used as the response variable. The predictors were the GPA at a given term and whether a student did an IUT transfer during their college career. This model allows for an examination of how the effects of GPA on the probability of a student switching majors within the CEAS and may be used to determine how this effect varied depending on what specific engineering major.

$$\Pr(y_i = 1) = \mathrm{logit}^{-1}(\beta_{j[i]1} G_i + \beta_{j[i]2} T_i + \beta_{j[i]3} G_i T_i + \alpha^{S_i}_{k[i]}) \tag{1}$$

The student's term GPA at term i was denoted by G_i, the students' number of terms enrolled was T_i. The slope coefficients $\beta_{j[i]1}$ and $\beta_{j[i]2}$ for the students' term GPAs and the number of terms a student was enrolled in respectively were used as a random effect to account for potential variations in this variable depending on how many terms a student had completed up to that point. An interaction between the term GPA and the number of terms was used to compare differences in slope for the term GPA depending on the number of terms enrolled in, its slope coefficient being denoted as $\beta_{j[i]3}$.

A simplified model similar to the one above was also developed, only including term GPA G_i as the sole fixed effect with no interaction term.

$$\Pr(y_i = 1) = \mathrm{logit}^{-1}(\beta_{j[i]1} G_i + \alpha^{S_i}_{k[i]}) \tag{2}$$

The simpler model was created to be used to create interpretable plots for the effects of term GPA by major in the CEAS.

D. Assumptions

All of the modeling performed in this analysis was constructed under the assumption that the curriculums and standards remained the same for all students throughout the course of the data found in the data set. Considering that this data spans 12 years, it is unlikely that the standards and curriculums of all programs within the CEAS remained exactly the same over that period of time. In order to make generalizations about movement patterns throughout the college, curriculum changes were mostly ignored throughout the analysis. In order to somewhat account for standards and curriculum changes, the data was broken up by cohort or starting year. Large pattern changes between cohorts would help to point out points in time in which curriculums or standards changed; for example, the data shows a dramatic decrease in the reporting of SAT scores beginning in 2021, corresponding to new university policies that year which no longer required SAT score submission in the wake of the COVID pandemic. In general, the analysis of these large pattern changes would be mostly left to people in charge of courses and with knowledge of course histories.

Furthermore, the data set provided information for up to 12 academic terms per student. Therefore, we assumed that each student's academic record was limited to 12 terms. A small number of cases suggested that some students may have taken additional terms beyond this window, however these instances were rare and unlikely to affect the analysis.

III. RESULTS

The following section provides an overview of the visualization tools created for the CEAS and their application to questions currently being asked by CEAS administrators. Additionally, hierarchical logistic regression models are fit to model the probability of a student switching their major under various conditions, and the analysis of these models is presented.

A. Visualization

Through the creation of different plots and visualizations of the data, the data confirmed many standing hypothesis of the CEAS. Through the use of advanced visualization techniques and the creation of interactive visualizations that filter through data in real time, it is possible to see the flow between majors throughout time at CU Boulder.

The visualizations created in this analysis have proven to be an improvement over what is currently being used by the CEAS to model and predict flow between majors. Based on the provided data, we are able to clearly see the patterns in which students transfer in and out of majors. These findings will help departments prepare for large influxes or exoduses of students and better plan for class enrollments.

1. Department Growth and Retention

Several plots were created to show the proportions and number of students switching majors within the CEAS and majors in other colleges at CU Boulder.

An alluvium plot was created to represent the total number of students who graduated from each engineering department since 2013. Each plot depicts the number of students who were enrolled in each program throughout each term they were enrolled at CU Boulder and meant to represent growth of each department each semester. The goal of these plots was to determine when students who graduated from a certain program joined that program to determine when each program would receive the largest increase in enrollment. An example of one of these plots is Figure 1. The information that this plot conveys is that in the first semester (the bar represented by Major_01) of those who graduated with a degree in Mechanical Engineering (MCEN), about half were declared as Mechanical Engineering, a large chunk were non-engineering majors (NON_ENGR), another substantial chunk were Open Option Engineering (XXEN) and the rest were spread across all of the other engineering majors. Moving to the second term (Major_02), out of those same students from the first bar, more were now declared Mechanical Engineering and noticeably less were non-engineering majors and Open Option Engineering suggesting that the ones who transferred into Mechanical Engineering came mostly from non-engineering majors or Open Option Engineering. This shift in numbers from the first to second semesters gives some indication of when students who will eventually graduate from Mechanical

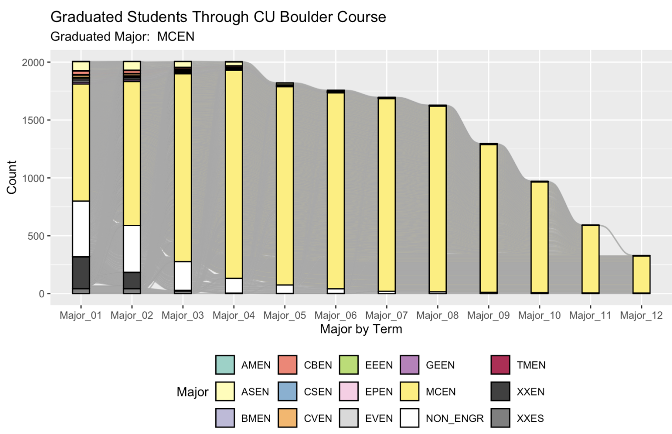

FIG. 1. Common pathways for students who graduated with a mechanical engineering degree.

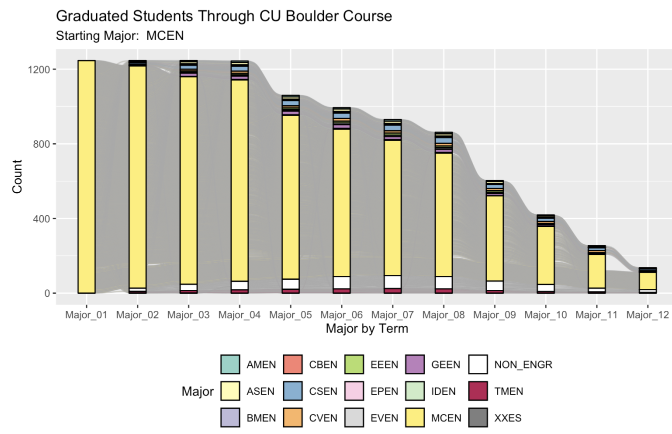

FIG. 2. Retention of students who were declared as Mechanical Engineering majors their first semester at CU Boulder.

Engineering transfer and settle into the program. This convention remains the same for the rest of the bars in the plot. This plot also gives indication into the times that students tend to take to graduate with a Mechanical Engineering degree. Most take eight semesters, but the data suggest that there are some that take longer than that, around an eighth of those who graduate take twelve semesters.

Another set of alluvium plots were made for each department. These plots were filtered down to represent students who graduated from CU Boulder and were enrolled in each engineering department during their first semester at CU Boulder. These alluvium plots were made to represent the retention of each department throughout the course of a student's time at CU. More specifically, it helped to show when and where students transferred when they left a specific program and the number of students who stayed in a given program from their first semester. These plots highlighted interesting findings. It sheds light on the fact that smaller programs such as Engineering Physics (EPEN) or Applied Mathematics Engineering (AMEN) tend to see a very large rate of students transferring out of the programs. In these cases, they tended to transfer out of engineering entirely. This was largely not the case for larger programs such as Aerospace (ASEN) or Mechanical (MCEN).

An example of this other kind of alluvium plot is Figure 2. The first bar represents the count of students whose first major at CU Boulder was Mechanical Engineering. Moving into their second semester (Major_02), there was little change in the number who were declared as Mechanical Engineering. Moving forward into the following semesters, it is clear that students tended to transfer mostly into other engineering majors but Mechanical Engineering tended to retain majority of its students as time went on.

Both sets of alluvium plots also shed light on interesting common paths. There tended to be a strong link between somewhat related engineering fields. Many students who left a particular engineering department tended to switch to a more related department. For example, the most common engineering major for students to declare when switching out of Computer Science was Creative Technology and Design, a user and experience centered major.

2. *Temporal Distribution of Switching*

Much of this analysis focuses on identifying common trajectories between departments that students switch between. From an administrative standpoint, it is also helpful to know how many semesters these students have completed in their origin program before they switch their major. This is particularly relevant for the CEAS, as students with multiple completed semesters in engineering majors likely have more background knowledge, which informs the allocation of resources across sophmore, junior, and senior level courses. To elucidate the distribution of the number of semesters completed for students who switch majors, bar charts are used, and an example can be found in Figure 3. This section will detail the functionality of this visualization tool, first describing the selection of basic plot features, followed by discussion of data filtering options. Note that details of all user-selected features are described in the integrated labels.

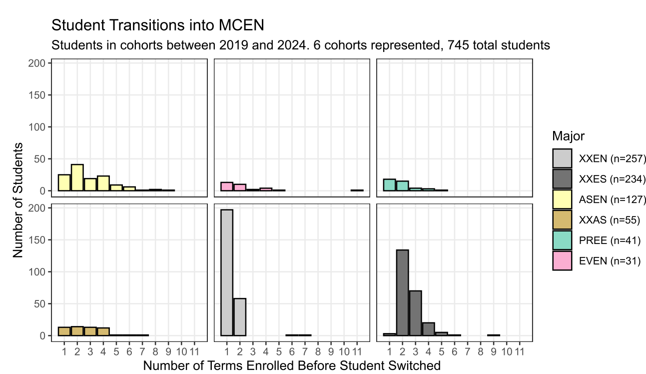

FIG. 3. Visualization app bar-chart demonstration, showing student transfers into MCEN.

Bar charts display the number of observations that fall within a certain discrete category. For the purposes of the following plots, the x-axis of the bar chart corresponds to the number of terms a student has completed, either in the CEAS or at CU in general, according to user input. Each bar will then represent the number of students who made a particular transition between two majors after completing a certain number of terms.

To select the majors whose transitions will be represented, the user must first specify a primary major of interest. The user will also specify if they want to visualize transitions into or out of the major of interest. In Figure 3, the major of interest is Mechanical Engineering (MCEN), and the plot shows transitions into this major. The app will then internally rank all other majors according to the magnitude of observed student transitions between it and the major of interest, after applying various user-specified data filters. The user may select the number of secondary majors to consider, and that many majors will be selected according to the largest magnitude. The app will then create a series of bar charts, with each bar chart representing transitions between the major of interest and one of the secondary majors, with the secondary major represented by color.

The user must choose an interval of student cohorts from which to subset the data, with cohort being defined as the year of the student's first term in the CEAS. An example of this functionality with a different major of interest, direction, and cohort selection can be found in Figure 4 which displays the paths that students in cohorts between 2015 and 2020 take when transferring out of Computer Science (CSEN).

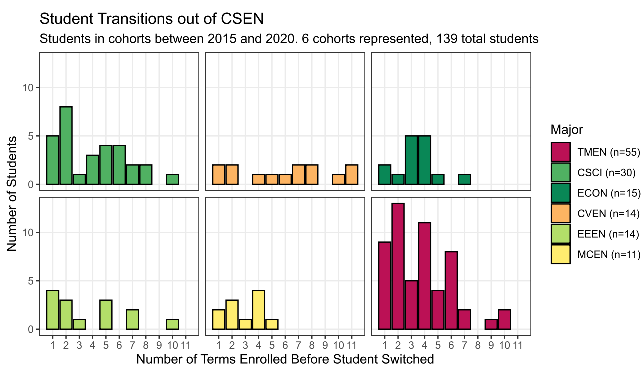

FIG. 4. Visualizaiton app bar-chart demonstration, showing transfers out of CSEN for 2015-2020 cohorts

Additional filter options concern the behavior of exploratory students, who often show different major switching behavior. The user may choose to leave these majors in the data, select only exploratory programs, remove exploratory programs, or collapse all exploratory major codes into one code "EXPL". Figure 5 demonstrates the option to select only exploratory programs, as well as an option to standardize the scale of each chart. Raw counts can be found in the color legend.

An additional argument allows for students who completed an IUT transfer to be specifically selected and visualized, demonstrated in Figure 6. Note that exploratory students are also excluded in this plot.

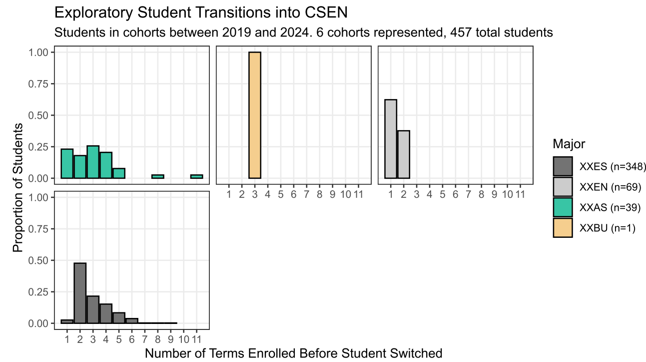

FIG. 5. Demonstration of selection of exploratory student transitions, as well as option for standardization of Y axis.

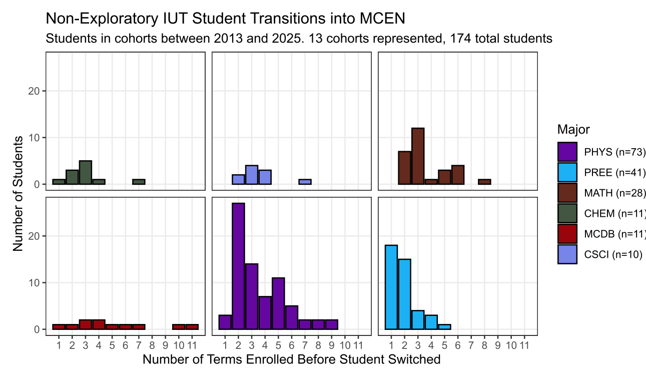

FIG. 6. Demonstration of selected IUT student transitions into MCEN, excluding exploratory programs.

3. *Flow Dynamics by Term*

Network diagrams can be used to visualize the flow in and out of several categories. For this project, an program was designed to create interactive directed network diagrams displaying the flow of students in and out of a CEAS Majors.

The user can select from many options to produce a specific diagram relating to specific majors and years. These include options to choose a specific year or range of years of entry into CEAS, the term of the students, a specific major to focus on, and any range of Term or Cumulative GPAs of the students.

For Example, Figure 7 displays the Diagram with these specifications: Cohort of 2019, Term Four, Major of MCEN, and any GPA between zero and four. This diagram shows all students who entered, stayed in, or left the mechanical engineering program in their fourth term who entered the CAES in the year 2019.

The circles in the diagram represent the majors, and the arrows represent the flow between these majors. Thus one can ascertain that in this term, 215 students stayed in the MCEN major, four students transfered from MCEN to CSEN, one student trnasfered from CSEN TO MCEN, three students transfered from ASEN to MCEN, etc.

Figure 8 shows the network diagnostic with the specifications: Cohort range of 2019 to 2022, Term Three, Major of

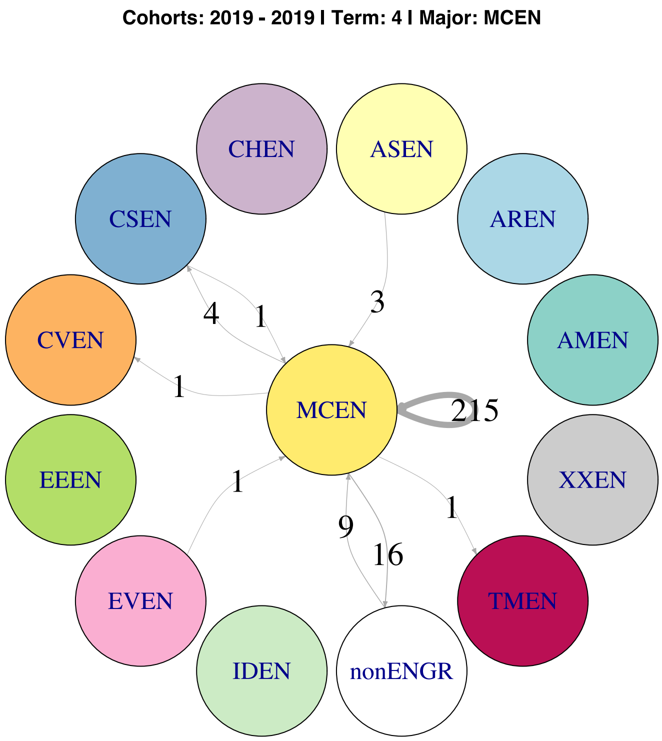

FIG. 7. An example of a network diagram

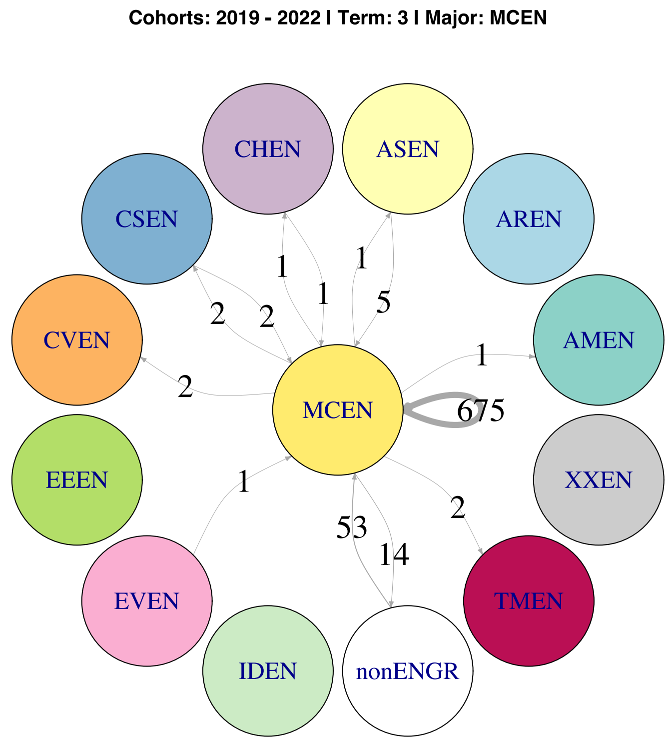

FIG. 8. Another example of a network diagram

MCEN, and GPA range of three and four.

4. *GPA Trends*

To better understand how academic performance relates to movement between engineering majors, a series of bar plots was generated. First, a major difficulty ranking was constructed by computing each student's mean GPA within a major and then averaging these values between students, retaining only majors with at least 30 students. Figure 9 displays the resulting average GPA across engineering majors in CEAS. This visualization highlights the overall GPA distribution, so we can clearly see each major relative to the others.

The results indicate that TMEN has the highest mean GPA, at approximately 3.4. Majors such as AMEN, CSEN, and AREN also maintain higher average GPAs (around 3.25 or above). In contrast, programs such as CBEN, ASEN, and CVEN tend to fall on the lower end of the distribution, generally below 3.1. Although the differences across majors are not extreme, they form a noticeable gradient in academic performance. This gradient is valuable when interpreting major-switching patterns, particularly in assessing whether students tend to move toward majors with higher or lower average GPAs.

Figure 10 highlights the most common pathways within the engineering department, ordered from the most frequent major switches at the top to the least frequent at the bottom (restricted to transitions made by more than 40 students). The chart also displays the average GPA differences associated with each transition. Positive bars indicate a switch into a major with a higher average GPA, while negative bars indicate a move into a major with a lower average GPA.

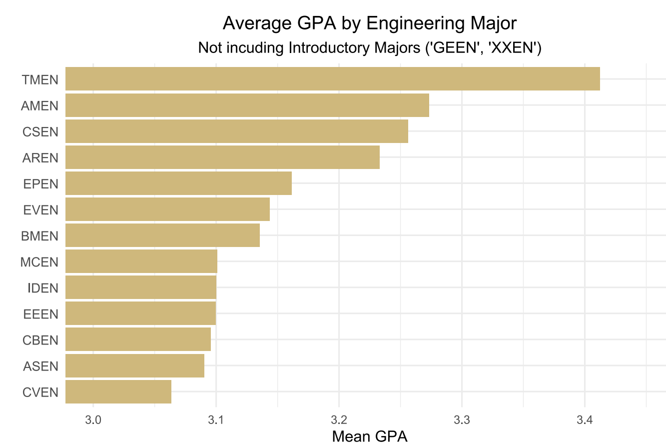

FIG. 9. Average GPAs of CEAS Majors

At the top of the chart, the most common major switch is from Aerospace Engineering to Mechanical Engineering, a transition that shows a negligible difference in average GPA. The predominance of positive bars indicates a general trend in which students move from lower-GPA majors into higher-GPA ones. Notably, CSEN, TMEN, and MCEN are among the most common destinations for switching students, while ASEN and CBEN experience the greatest attrition. As shown in Figure 9, ASEN and CBEN also rank among the lower-performing majors in terms of average GPA, which helps contextualize these outflows. Taken together, this visualization provides clearer context for interpreting student switching behavior by considering both the magnitude of GPA differences and the number of students involved in each pathway.

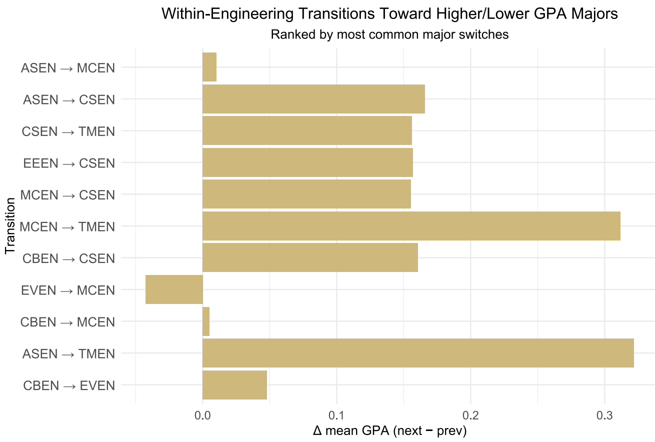

FIG. 10. Change in Average GPA Associated with the Most Common Transitions Within Engineering

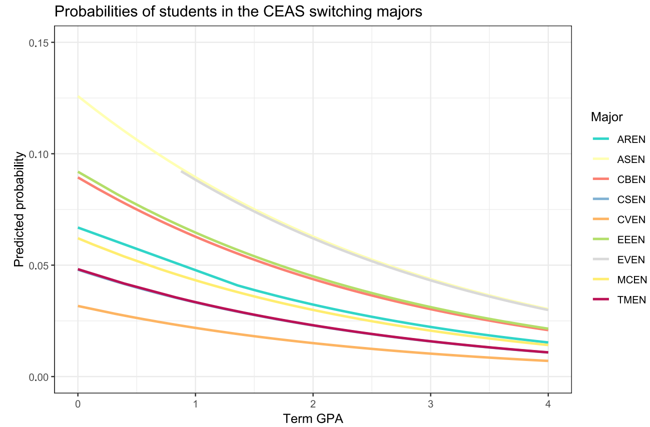

FIG. 11. The probability of students in the CEAS switching majors

B. Modeling

This subsection discusses the results and inferences produced from the multilevel logistic regression model as described in Section II C 2.

1. Findings

The model as shown in equation 1 had a slope coefficient estimate for the term GPA $\beta_{j[i]1}$ of -0.05876, for the number of terms enrolled $\beta_{j[i]2}$ of 0.01864, and the interaction between the two variables previously mentioned $\beta_{j[i]3}$ -0.08359. These three coefficients had standard errors of 0.07310, 0.05179, and 0.01643 respectively and p-values of 0.422, 0.719, and $3.6 \cdot 10^{-7}$ respectively. Additionally, the model had an Akaike information criterion (AIC) of 9884.7.

Only the coefficient estimate for the interaction was found to be statistically significant at a significance level of $\alpha = 0.05$. Generally, term GPA had a negative relationship with the probability of switching majors, while the number of terms enrolled increased that probability. However, their interaction shows that the probability of switching decreases with higher term GPAs and a greater number of terms enrolled, indicating the term GPA had the effect of lessening the probability that may be increased by a students' number of terms.

2. Plots on Probability Inference

The effect of term GPA on the probability of students switching majors within the CEAS will be examined with a line plot. Using the multilevel logistic regression model previously denoted in equation 2, the predicted probability of switching was calculated. This was done on a subset the data set only including students who stayed in the CEAS during their entire time at CU.

There is some variation of the effect of term GPA on switching majors depending on the major which can be seen in Figure 11. The effect appears to be lower for computer science majors and is noticeably higher for electrical and aerospace engineers. In particular with the latter two, the probability of switching remains higher compared to other majors even with a higher term GPA. The same can be said, to a lesser extent, with electrical and chemical engineering. In contrast, civil engineering, even at lower term GPAs, appear to have the lowest probability of switching.

IV. DISCUSSION

In our study, we developed a model from which we were able to identify trends in switching behavior between and within specific majors within the CEAS. In addition to these characterizations of broader switching behavior, visualization software is created which allows for administrators to more effectively parse patterns of student transitions in relation to specifically chosen programs.

A. Significance

With much improved visualizations that are interpretable and interactive, administrators and course planners at the CEAS will be better able to understand the dynamics of major switches within the college. The creation of an app gives administrators greater control of which subsets of the data set are displayed in the plots it produces. This enables a deeper exploration of various trends limited to specific cohorts, terms, majors, and other variables that previously could not be addressed from previous visualizations or publications.

Furthermore, the app's focus on transitions relative to a specific CEAS major are valuable in identifying which major-to-major transition is most common (e.g., from Aerospace

Engineering to Mechanical Engineering), and will be useful for examining departmental growth and allowing administrators to better allocate resources. The integration of term-level information allows for further characterization of the academic background students carry when they switch, which is valuable for anticipating the background knowledge of incoming transfers.

In addition, a model may allow for a better understanding of the various effects of variables such as term GPA on the probability of students switching majors. The results of the model are useful for analyzing the relative probabilities of students in engineering majors switching to other engineering majors. This can help give administrators and course planners better intuition as to the dynamics of major switching, which may be useful for determining course difficulty and resource allocation for specific courses.

B. Limitations

Our app has several limitations. Our data included only the students enrolled in the CEAS from the fall of 2013 through the fall of 2025, and the app currently does not include a method to incorporate future enrollment data. This means that the relevance of the visualizations will diminish as time goes on without manual adjustment. Another limitation of the app is that it is not integrated with CU Boulder's current data access infrastructure. This means there will be more friction for CEAS administrators to use the app and thus it might not be utilized to its full potential.

Our model also had several limitations. Because the data represented all students who, at some point, were enrolled in the CEAS, students who started in non-CEAS colleges were not used. Therefore, the data used for the model was a narrow subset of the data set provided, only including students who were enrolled only in the CEAS during their entire matriculation at CU Boulder, and who graduated with a degree in an engineering major. This subset excludes students who have dropped out or who have done an IUT transfer to or from the CEAS. Because of this, the model is unable to determine the probabilities of IUT students switching majors within the CEAS, a crucial element in advisor intervention. Furthermore, the model included only two predictors, which significantly limited prediction accuracy, essentially limiting the model to be inferential rather than predictive. The model is only further limited by the non-linear relationship the predictors have with the logit probability.

C. Conclusions

The goal of this project was to develop an effective and interpretable workflow that helps CU Boulder's College of Engineering and Applied Science understand and anticipate changes in major enrollment. Through the development of interactive visualizations, we were able to clearly depict term-by-term population flow, highlighting patterns in how students progressed through various programs.

Across these tools, consistent patterns emerged: exploratory programs and certain engineering majors act as major entry points into others, switching tends to cluster at specific terms within a student's academic timeline, and many transitions align with GPA differences between majors. These visuals are presented in a format that is both accessible and interactive, allowing for the application of these findings to a broad scope of future questions CEAS administrators may hope to answer.

The visual framework created through this paper enables CEAS administrators to better anticipate fluctuations in program sizes, allocate resources more effectively, and design targeted retention efforts.

D. Recommendations

For future development of the app, there are many improvements that could be developed. Our limitations display some of these improvements that we would have liked to make. A feature that allows a user to input new data would greatly improve the lifespan of the app. It would also greatly improve the ease to use the app if we had worked with CU Boulder's data access team to integrate it into the CU Boulder's data portal. Additionally, user testing could help refine the user interface and usability of the app.

Concerning the model, we would first recommend including more predictors in the data set. Doing this would not only improve model accuracy but also reveal trends in different groups at the CEAS. Some predictors that could be useful are demographic information such as gender and ethnicity. However, we stress that approaching this must be done with great caution so that this data is used in an ethical manner. By including sensitive demographic data, those with access to the data set could potentially find personally identifiable information. Next, we recommend developing a model that considers the non-linear relationship the predictors had in our model with the response. This would create a model that can more accurately predict a student switching majors based on those variables and may be helpful for administrators and advisors to better allocate resources and help students with interventions. Finally, we recommend to expand the model to include a wider range of students in the CEAS, including those who did an IUT transfer. This will overcome the limited scope of the model's analysis and enable a better understanding the probability of switching for those populations excluded in our model's data subset. From this, administrators and advisors can devise better intervention strategies for those CEAS student populations.

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V. APPENDIX

Official Degree Name	Engineering Code	Analysis Code
Aerospace Engineering	ASEN	ASEN
Architectural Engineering	AREN	CVEN
Applied Mathematics Engineering	AMEN	AMEN
Biomedical Engineering	BMEN	BMEN
Biological and Chemical Engineering	CBEN	CBEN
Biological Engineering	BIEN	CBEN
Chemical Engineering	CHEN	CBEN
Civil Engineering	CVEN	CVEN
Computer Science Engineering	CSEN	CSEN
Computer Science Bachelor of Arts	CSBA	CSEN
Electrical Engineering	EEEN	EEEN
Electrical and Computer Engineering	ECEN	EEEN
Engineering Physics	EPEN	EPEN
Environmental Engineering	EVEN	EVEN
Integrated Design Engineering	IDEN	IDEN
Mechanical Engineering	MCEN	MCEN
Creative Technology and Design	TMEN	TMEN